THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET C

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Which nutrient do insectivorous plants get from insects? 1

(a) oxygen **(b)** carbon **(c)** calcium **(d)** nitrogen

2. Prepared food is transported to all parts by the: 1

(a) phloem **(b)** stomata **(c)** root hairs **(d)** xylem

3. The upper surface of a leaf is usually: 1

(a) darker green and receives more light **(b)** full of roots
(c) without chlorophyll **(d)** white

4. Which gas is released during photosynthesis? 1

(a) oxygen **(b)** hydrogen **(c)** nitrogen **(d)** carbon dioxide

5. Materials that allow light to pass completely are: 1

(a) luminous **(b)** translucent **(c)** transparent **(d)** opaque

6. A rough surface causes: 1

(a) regular reflection **(b)** clear images
(c) no light **(d)** diffused reflection

7. The bouncing back of light from a surface is: 1

(a) dispersion **(b)** refraction **(c)** absorption **(d)** reflection

8. Image in a pinhole camera is ___ if the hole is large. 1

(a) sharp **(b)** brighter only **(c)** smaller **(d)** blurred

9. Which causes the apparent movement of the Sun across the sky? 1

(a) wind **(b)** Earth's revolution
(c) Moon's revolution **(d)** Earth's rotation

10. The pole star appears nearly fixed because it lies: 1

(a) near the extension of Earth's axis **(b)** at the equator
(c) near the Moon **(d)** beside the Sun

11. The cycle of Moon phases repeats in about: 1

(a) 24 hours **(b)** 29.5 days **(c)** 7 days **(d)** 365 days

12. The Moon shines because it: 1

(a) reflects sunlight **(b)** burns
(c) is hot **(d)** makes light

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 15–25 words each.

13. Why is transpiration useful to plants? 2

14. What are stomata? Give one function. 2

15. State the law of reflection. 2

16. Differentiate luminous and non-luminous objects with examples. 2

17. What is rotation of the Earth? What does it cause? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. How do legumes improve soil fertility? 3

19. Explain how a shadow is formed. 3

20. State and explain the characteristics of an image in a plane mirror. 3

21. Differentiate solar and lunar eclipses. 3

22. Explain the formation of a solar eclipse. 3

OR

Explain why the night sky looks different at different times of the year. 3

23. Why is photosynthesis important for life on Earth? 3

OR

Describe the modes of nutrition other than photosynthesis in plants. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

24. (i) Define photosynthesis. 5
(ii) Name the raw materials and products.
(iii) Where does it occur?

OR

Case study: Ravi finds a plant with no leaves twined around a tree. 5
(i) Name it.
(ii) Mode of nutrition.
(iii) Effect on host.
(iv) Another mode of nutrition in plants.

25. **Case study:** A girl holds a card with 'ROSE' in front of a mirror. 5
(i) How does it appear?
(ii) Name the effect.
(iii) Distance of image if card is 40 cm away.
(iv) Nature of image.

OR

(i) Describe the construction of a pinhole camera. 5
(ii) State the nature of its image.
(iii) Effect of hole size.

END OF QUESTION PAPER — Total: 50 Marks